DATA ANALYTICS PORTFOLIO PROJECT

CUSTOMER ANALYTICS & RFM SEGMENTATION

IKEA Gifting | End-to-End Analytics & Reporting Documentation

Technical stack	SQL Server, Python (Pandas, NumPy), Power BI
Dataset	Online Retail II transaction dataset (>1,000,000 records)
Analytical focus	Customer purchasing behaviour, RFM segmentation, repeat-customer revenue and product revenue concentration
Project structure	Tutorial-guided RFM core + two independently authored SQL extensions

Executive Summary

This project analyses more than one million historical retail transaction records using SQL Server, Python, and Power BI. The work combines transaction data preparation, customer-level Recency, Frequency and Monetary (RFM) analysis, customer segmentation, and interactive reporting. The dashboard summarises 5,881 identified customers and approximately $17.4 million in reported revenue. Two additional SQL analyses independently examine the revenue contribution of repeat customers and the concentration of revenue among the top 10 products.

The core RFM workflow and initial dashboard design were developed through a tutorial-guided learning process and then customized and recreated. The two SQL extensions were independently authored as additional analysis.

> **Case-study clarification:** "IKEA Gifting" is the project case-study name. The analysis uses the publicly available Online Retail II dataset; it is not actual IKEA company data and is not affiliated with IKEA.

1. Business Context & Objectives

In retail e-commerce, understanding customer purchasing behaviour, repeat purchases, and product revenue concentration can support better business decisions. This project explores historical transaction patterns from an online retail dataset and converts raw order records into customer and product performance indicators.

Project objectives:

- Quantify customer value through purchase recency, purchase frequency, and total spend.
- Apply RFM segmentation to classify customers into Champions, Loyal, At Risk, Lost, New, and Needs Attention groups.
- Compare customer and revenue distribution across selected geographic markets.
- Independently analyse repeat-customer revenue contribution and top-product revenue concentration using SQL.

2. Tools & Data Workflow

Stage	Tool	Work performed
Data preparation	SQL Server / SSMS	Imported and combined transaction tables and prepared the consolidated transaction dataset for analysis.
RFM analysis	Python: Pandas, NumPy	Filtered records with non-null Customer_ID and positive Quantity; calculated customer-level RFM metrics, assigned 1-5 scores, and applied segment rules.
Independent extensions	SQL Server / SSMS	Compared repeat and one-time customers; separately calculated top-10 product revenue concentration, excluding StockCodes M and POST for that product analysis.
Dashboard reporting	Power BI	Built interactive report pages, KPI measures, slicers, cross-filtering visuals, page navigation, and a Key Metrics Glossary.

3. Contribution & Methodology Disclosure

The core RFM calculation workflow and initial Power BI dashboard design followed a tutorial-guided learning structure and were subsequently customized and recreated. The Repeat Customer vs. One-Time Customer Comparison and Top 10 Product Revenue Concentration analyses were independently authored in SQL as original extensions to the guided core project.

This distinction identifies the guided components transparently while highlighting the additional analysis completed independently.

4. RFM Methodology

The RFM model evaluates customers across three purchasing dimensions. The Python workflow reads the cleaned transaction file, converts InvoiceDate to datetime, filters eligible transactions, aggregates customer-level metrics, assigns quantile-based scores, and merges the customer metrics back onto transaction-level records for dashboard reporting.

Dimension	Definition used in this project
Recency (R)	Number of days between a customer's most recent transaction and the dataset reference date.
Frequency (F)	Number of distinct invoices/orders placed by the customer.
Monetary (M)	Total customer spending, calculated by summing Quantity × Price across included transactions.

RFM processing rules confirmed from the Python code:

- **Eligible rows:** Customer_ID is not null and Quantity is greater than zero.
- **Recency reference date:** 10 December 2011.
- **Recency score:** Reverse-ordered, so 5 represents the most recent purchasing behaviour and 1 the least recent.
- **Frequency and Monetary scores:** 1-5 quantile-based scores, with higher scores representing greater order frequency and spending.
- **Churn flag:** IsChurned is True when Recency is strictly greater than 90 days.

RFM segment assignment rules:

Segment	Rule in Python code	Interpretation
Champions	R ≥ 4, F ≥ 4, M ≥ 4	Recent purchasers with high frequency and high monetary value.
Loyal	R ≥ 3, F ≥ 3 (after Champions)	Relatively recent customers with higher purchase frequency.
At Risk	R ≤ 2, F ≥ 3	Less recent customers with relatively higher purchase frequency.
New	R ≥ 4, F ≤ 2	Recent purchasers with lower purchase frequency.
Needs Attention	R = 3, F ≤ 2	Customers with middle-range recency and lower purchase frequency.
Lost	Remaining customers (R ≤ 2, F ≤ 2)	Less recent customers with lower purchase frequency.

> **Churn interpretation:** The 90-day rule is a project-defined inactivity flag, not proof that a customer has permanently left the business. It is measured against the fixed dataset reference date of 10 December 2011.

5. Dashboard Overview & Performance Indicators

The Power BI report presents customer and revenue performance through interactive pages and slicers. The report structure includes Business Overview, RFM Analysis, and Customer & Product Insights. It also includes a Key Metrics Glossary to explain principal measures to end users.



Figure 1: Power BI Business Overview Dashboard showing overall metrics and segment breakdowns.

Metric	Reported value	Meaning
Total Customers	5,881	Identified customers included in the analysis.
Total Revenue	$17.4M	Total revenue shown in the Power BI overview.
Churned %	50.9%	Share of identified customers with Recency greater than 90 days.
Average Customer Value	$2,954	Total Revenue divided by Total Customers, representing the average revenue per identified customer.

Customer segment revenue breakdown:

Segment	Behavioural profile based on code	Revenue
Champions	Recent, high-frequency, high-spend customers.	$11,836,648
Loyal	Relatively recent customers with higher frequency.	$2,701,267
At Risk	Less recent customers with higher frequency.	$1,586,359
Lost	Less recent customers with lower frequency.	$653,885
New	Recent customers with lower frequency.	$392,022
Needs Attention	Middle-range recency and lower frequency.	$204,624
Total	**All six customer segments.**	**$17,374,805**

> **Key insight:** Champions account for approximately 68.1% of reported total revenue ($11,836,648 of $17,374,805), making them the largest revenue-contributing segment in this dataset.
> The churn percentage should be interpreted using the project's 90-day inactivity definition and 10 December 2011 reference date. It is a descriptive classification of the historical dataset.

6. Geographic Analysis

The RFM Analysis dashboard includes a country-level view of customer counts, revenue, and average RFM scores. The following figures are the selected-country summary presented in the project report.

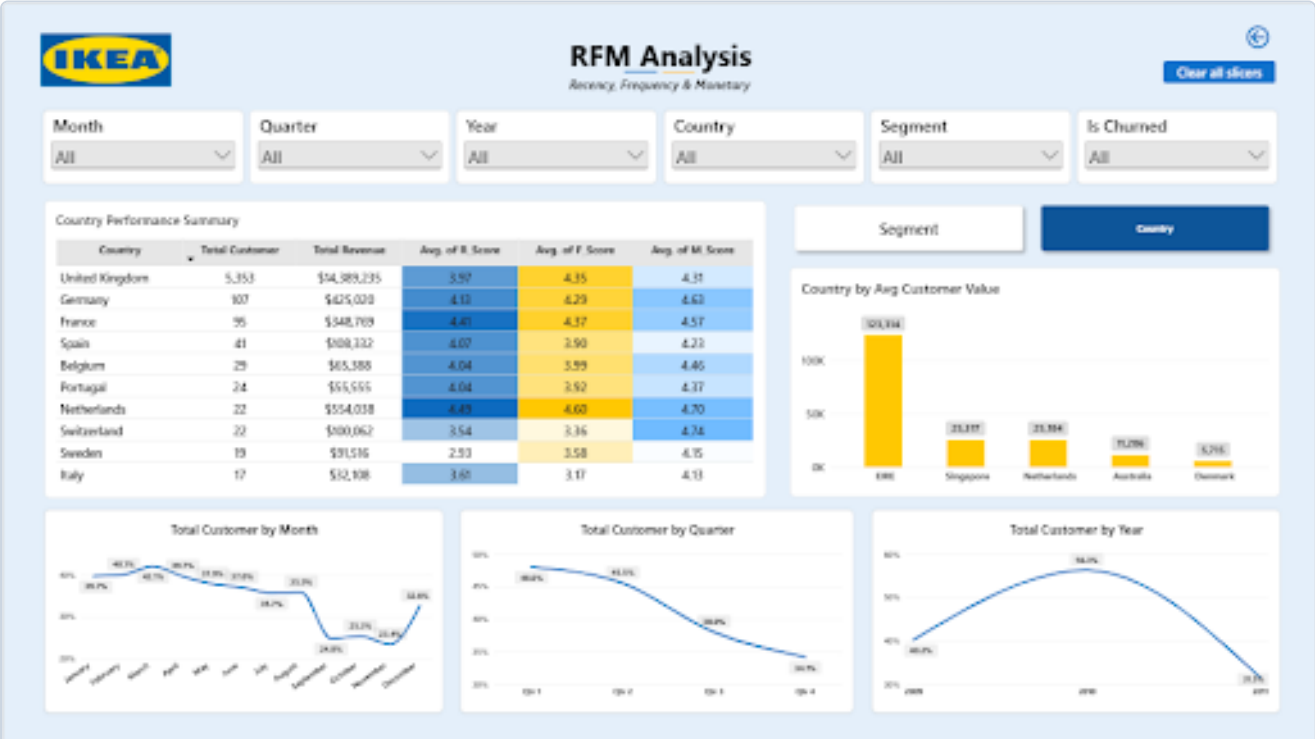


Figure 2: Power BI RFM Analysis Dashboard highlighting geographic breakdown and customer metrics.

Country	Customers	Revenue ($)	Avg R	Avg F	Avg M
United Kingdom	5,353	$14,389,235	3.97	4.35	4.31
Germany	107	$425,020	4.13	4.29	4.63
France	95	$348,769	4.41	4.37	4.57
Spain	41	$108,332	4.07	3.90	4.23
Netherlands	22	$554,038	4.49	4.60	4.70

Market observation:
The Netherlands shows a relatively high average revenue per identified customer ($25,184) in the selected-country summary, based on 22 customers. This is a descriptive observation from a small customer base. Further work on customer acquisition costs, market size, and purchasing trends would be needed before drawing market expansion conclusions.

Scope note:
The table presents selected countries rather than the complete geographic distribution. The five displayed customer counts sum to 5,618, compared with 5,881 identified customers overall; therefore, other countries are not represented in this selected summary.

Interpretation caution:
Average R, F, and M scores are summary measures for the selected country groups. They should be read alongside customer counts and revenue, rather than treated as standalone evidence of market opportunity.

6.1 Customer & Product Insights

This dashboard page provides additional insights into customer purchasing frequency, repeat-customer revenue contribution, and top-performing products. It brings together customer behaviour and product-level performance indicators in one view.

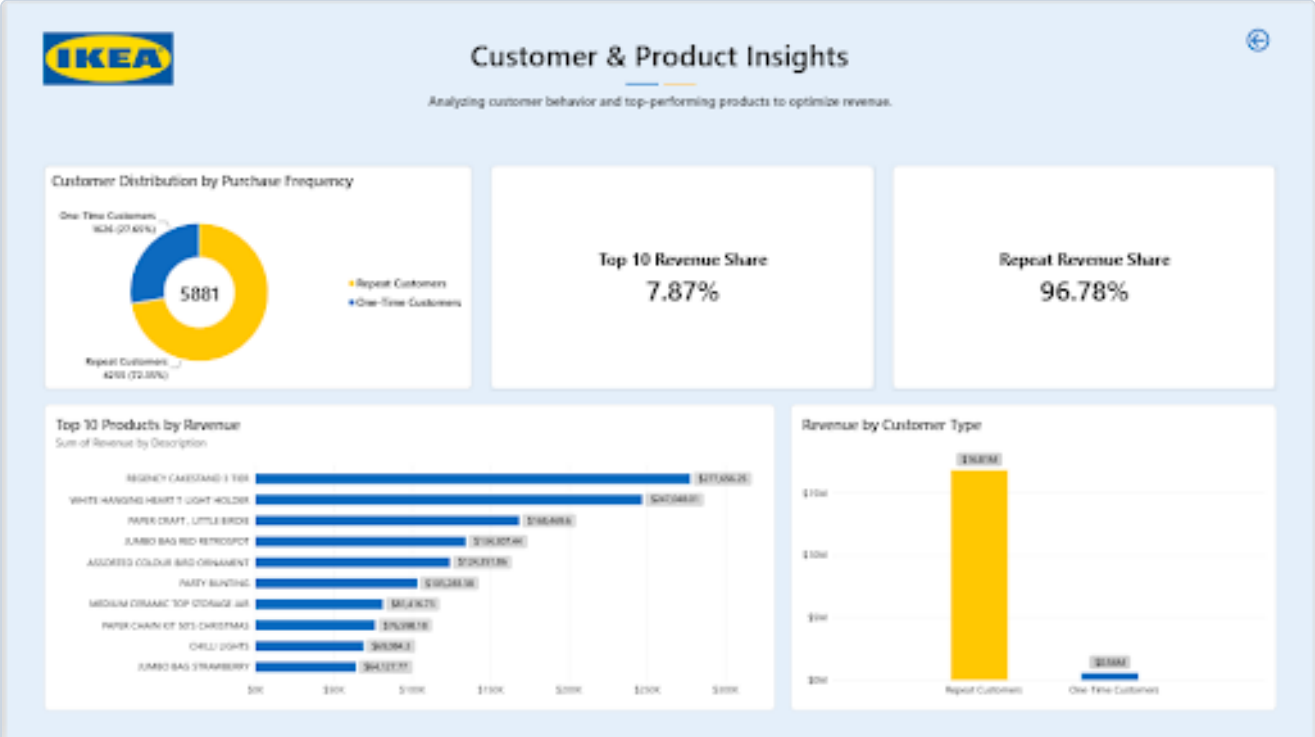


Figure 3: Power BI Customer & Product Insights Page showing top revenue products and customer repeat behaviour.

7. Independent SQL Extension 1: Repeat vs. One-Time Customers

Business question: How much revenue is generated by repeat customers compared with customers who placed only one order?

The analysis grouped eligible customers by the number of distinct invoices. Customers with more than one distinct invoice were classified as repeat customers; customers with exactly one distinct invoice were classified as one-time customers. The SQL analysis included records with a non-null Customer_ID and positive Quantity.

Customer group	Customers	Customer share	Revenue share	Group AOV ($)
One-time	1,626	27.65%	3.22%	$344.57
Repeat	4,255	72.35%	96.78%	$475.67

* Group AOV = Group Revenue / Group Order Count

Finding: Repeat customers contribute 96.78% of analysed revenue while representing 72.35% of identified customers. Their group-level Average Order Value is approximately 38% higher than that of one-time customers. This highlights the contribution of repeat purchasing and suggests that customer retention deserves further investigation.

8. Independent SQL Extension 2: Top 10 Product Revenue Concentration

Business question: What proportion of eligible product revenue is generated by the 10 products with the highest revenue?

The product-level analysis ranked eligible products by total revenue and compared the combined revenue of the top 10 products with total eligible product revenue. Records with a non-null Customer_ID and positive Quantity were included; StockCodes M and POST were additionally excluded from this product analysis as non-physical/service codes for this project.

Measure	Result
Top 10 products combined revenue	$1,346,343.52
Top 10 share of eligible product revenue	7.87%
Remaining product revenue	$15,752,035.02
Remaining share	92.13%
Highest-revenue product listed	Regency Cakestand 3 Tier ($277,656.25)

> **Methodology distinction:** Both SQL extensions use non-null Customer_ID and positive Quantity. The top-10 product analysis additionally excludes StockCodes M and POST. Its eligible revenue total therefore differs from the customer-type analysis revenue total.

9. Business Opportunities for Further Investigation

The findings are descriptive and can guide follow-up questions. They do not establish that a particular intervention will produce a specific business result.

- **Customer retention:** Explore targeted re-engagement campaigns for customers classified as At Risk, then measure subsequent purchase behaviour.
- **Loyalty initiatives:** Evaluate programmes intended to encourage repeat purchases and compare customer behaviour before and after implementation.
- **Product performance:** Review revenue and sales performance across the wider product catalogue before making inventory allocation decisions.

10. Dataset & Methodology Limitations

- **Historical dataset:** The analysis uses a static historical transaction dataset and does not predict future sales.
- **Missing customer attribution:** Transactions without a Customer_ID cannot be attributed to individual customers and are excluded from customer-level analysis.
- **Project-defined churn:** The 90-day inactivity threshold is a chosen analytical rule, not a verified business churn definition.
- **Descriptive analysis:** Historical purchasing patterns cannot establish whether retention campaigns or loyalty initiatives would improve revenue.
- **Product exclusions:** StockCodes M and POST are excluded only from the top-10 product concentration analysis; results are specific to that analysis scope.

11. Conclusion

This project demonstrates an end-to-end analytics workflow involving transaction data preparation, customer-level RFM segmentation, SQL-based business analysis, and interactive Power BI reporting. The tutorial-guided core workflow was extended through two independently developed SQL analyses examining repeat-customer revenue contribution and top-product revenue concentration. The findings provide a descriptive view of customer purchasing behaviour and product performance, highlighting areas for further business investigation.

Appendix: Key Metric Definitions

Metric	Definition
Churned %	Share of identified customers whose Recency is greater than 90 days.
Repeat Customer %	Share of identified customers with more than one distinct invoice.
One-Time Customer %	Share of identified customers with exactly one distinct invoice.
Top Revenue Share %	Top 10 product revenue divided by total eligible product revenue for the product analysis.

> Prepared as a portfolio case study. Reported figures are reproduced from the project details supplied.